

Entropy

Defining surprise / information

Surprise has an inverse relation with probability.

- High probability = low surprise (if an outcome is almost certain not surprised when it happens)
- Low probability = high surprise

⚠ Surprise can also be interpreted as amount of information.
Unlikely events contain more informations than usual events.

Mathematically: if $p(\text{event}) = 1$ surprise should be 0

$$\text{Therefore can be described with: } \text{Surprise} = \log_2\left(\frac{1}{p}\right) = \log_2(1) - \log_2(p) \\ = -\log_2(p)$$

Defining Entropy

Entropy is the average expected amount of surprise per event.

Example: entropy of flipping a coin tells how much surprise you encounter on average per flip.

$$\text{Entropy} = \sum (p_i \times \text{Surprise}(p_i)) = - \sum p_i \log_2(p_i)$$



Maximum entropy = fair distribution

$$\text{Entropy} = -[0.5 \times \log_2(0.5) + 0.5 \times \log_2(0.5)] \\ = -[0.5 \times (-1) + 0.5 \times (-1)] = 1$$



Low entropy = high imbalance

$$\text{Entropy} = -\left[\frac{3}{4} \times \log_2\left(\frac{3}{4}\right) + \frac{1}{4} \times \log_2\left(\frac{1}{4}\right) \right] \\ = -\left[\frac{3}{4} \times (-0.415) + \frac{1}{4} \times (-2) \right] \approx 0.244$$